JÖNKÖPING UNIVERSITY
School of Engineering

Kickoff

Introduction to Script Programming, 7,5 ECTS

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Patrick Riehmann

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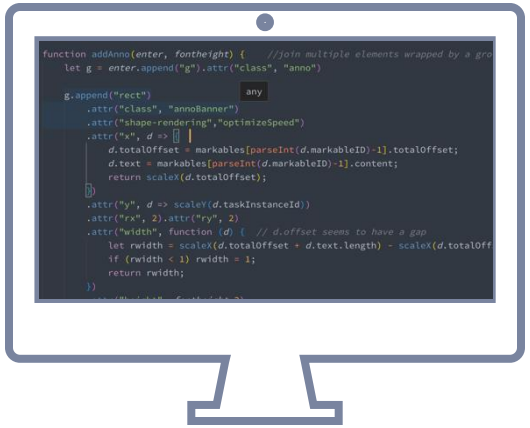
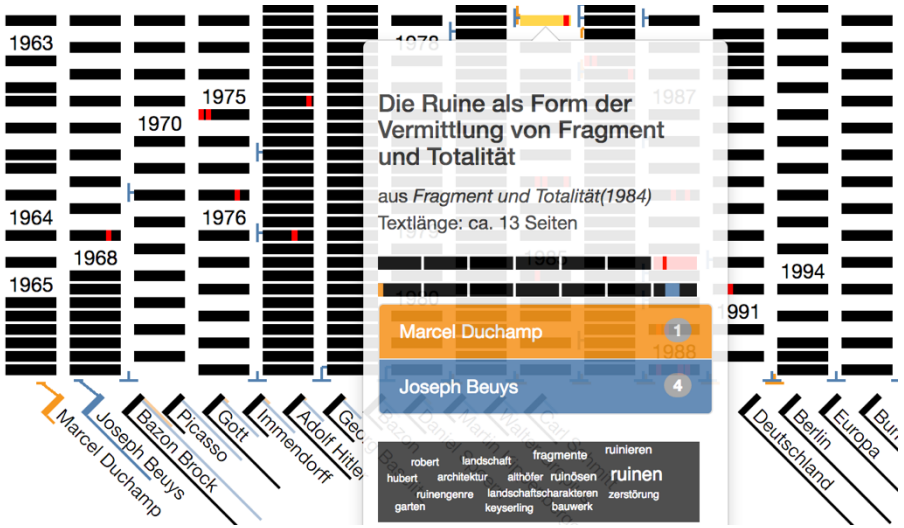
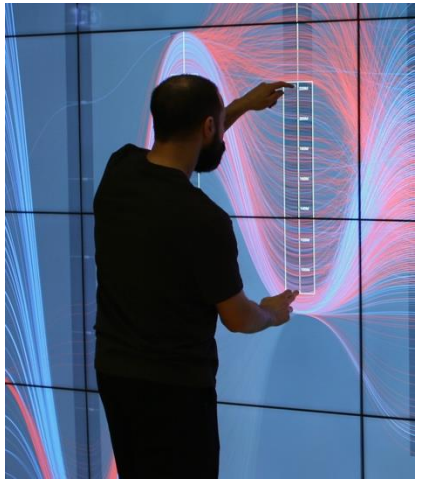


Software Development



Information Visualization



HCI

- Diploma in Computer Science and Media at Bauhaus-Universität Weimar, Germany
- PhD in Information Visualization and Visual Analytics
- With JTH since August 2023 as Assistant Professor Information Visualization

Web Technology

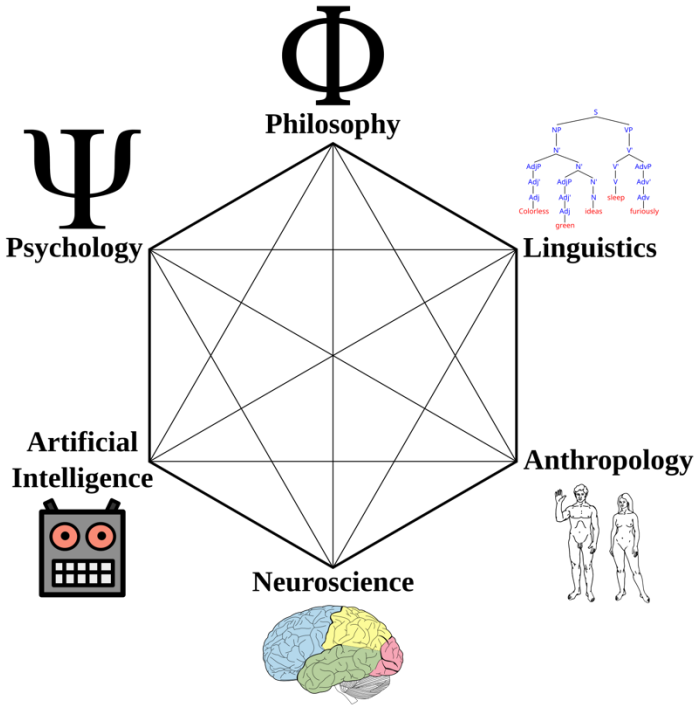
He Tan

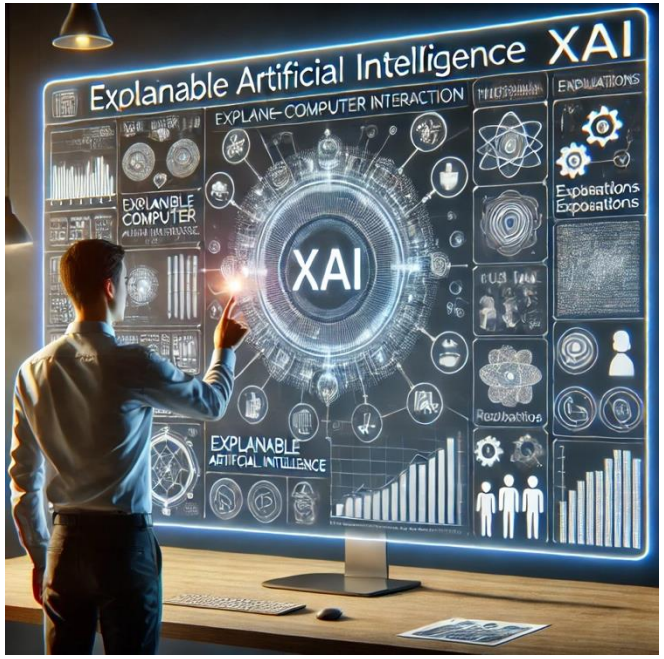
- Assistant Professor, Jönköping AI Lab, at the Department of Computing.
- Teaching: currently, course coordinator for the bachelor courses Data Structures and Databases, and the master course Knowledge Representation.
- Research:
  - Ontology, Knowledge Graphs, and their applications
  - Knowledge Representation and Reasoning
  - Symbolic AI

Linus Holmberg

Research Interests:

- Judgement & Decision Making
- Explanations (HCXAI)
- Human-Computer-Interaction
- Decision Support Systems





- Master in Cognition & Communication
- Master in Data Science
- PhD Student in Cognitive Science

General course information

- All Material
  https://pr-ju.github.io/course-material/courses/introduction-to-script-programming/

- Canvas for communication, submissions and points

- People
  - Patrick Riehmann
    (lecturer, course coordinator, examiner)
  - He Tan (lecturer, examiner)
  - Linus Holmberg (lecturer, examiner)

Program	Course Code
Sustainable Building Information Management	TSTG17
IT Infrastructure and Network Design	TSPG17
User Experience Design (electable)	
All other students (electable)	

Week	21. Oct 2024 1	28. Oct 2024 2	4. Nov 2024 3	11. Nov 2024 4	18. Nov 2024 5	25. Nov 2024 6	2. Dec 2024 7
Lectures							
L1	Basic 1	Functions	Conditions, Loops	Glyphs	Modelling	Abstractions	Q&A
L2	Basic 2	Modules	Strings	Dicts	Data Storage	Objects	Q&A
Lab Assigments							
Out	Lab 1 (Basic)	Lab 2 (Functions, Moduls)	Lab 3 (Conditions)	Lab 4 (Lists & Dicts)	Lab 5 (Working with Files)	Lab 6 (Objects)	
Due		Lab 1 (Basic)	Lab 2 (Functions, Moduls)	Lab 3 (Conditions)	Lab 4 (Lists & Dicts)	Lab 5 (Working with Files)	Lab 6 (Objects)

- Due means it must be uploaded in Canvas and one of the teachers must have had a look at it before and admitted it !!!
- This means presenting it in your respective lab class or at the latest on the last lab class in the respective week !!!
- Passing a lab means that everything can be run and behaves as the expected !!!

The Exam

- Made in Inspera
- Take place on the ?th of December
- Between ? – ?
- Consist of a lot of questions
- Lasts for 4 hours
- Grades
  - 50%, 70%, 90%

If everyone passes the exam, it makes us look good!

- Please pass the exam.

- No pressure ☺

- **Studēre**:

1. (*with dative*) to dedicate oneself,
   direct one's efforts or attention (to), strive after, be devoted to

2. to aspire to, tend to, desire, look for

3. (*with dative*) to be attached or favorable to, favor, support,

4. (*especially Late Latin*, *Medieval Latin*) to study, to apply oneself to learning

5. (*Medieval Latin*) to care,
   to think Synonyms: cōnsulō, cūrō, cōnsultō, colō, respiciō

What is a script?

- Wikipedia: https://en.wikipedia.org/wiki/Scripting_language

A **scripting language** or **script language** is a programming language that is used to manipulate, customize, and automate the facilities of an existing system. Scripting languages are usually interpreted at runtime rather than compiled.

- Wikipedia: https://en.wikipedia.org/wiki/Compiler

A **compiler** translates computer code written in one programming language into another language. It is primarily used for translating source code from a high-level programming language to a low-level programming language (e.g. machine code) to create an executable program

➢ Scripts do not run on their own! They need an interpreter!

What is a script?

- Scripts are human-readable text files with commands!
- Scripts do not run on their own!
- They need an interpreter program that executes those commands!
- The interpreter needs to run first!
- Interpreters could be:
  - Shells like powershell, bash, zsh
  - Web Browsers
  - Other programs such as graphics and video programs, etc.
  - Dedicated scripting languages: pearl, lisp, lua, ruby and python

Why Python? Is it any good?

g+ PYTHON

The second best programming language for everything

Python is one of the most used programming languages and very popular with many developers - and rightly so. But what actually makes her so special?

From Mike Müller

25. June 2023, 10:00 am

https://www.golem.de/news/python-die-zweitbeste-programmiersprache-fuer-alles-2306-175165.html

Why Python? Is it any good?

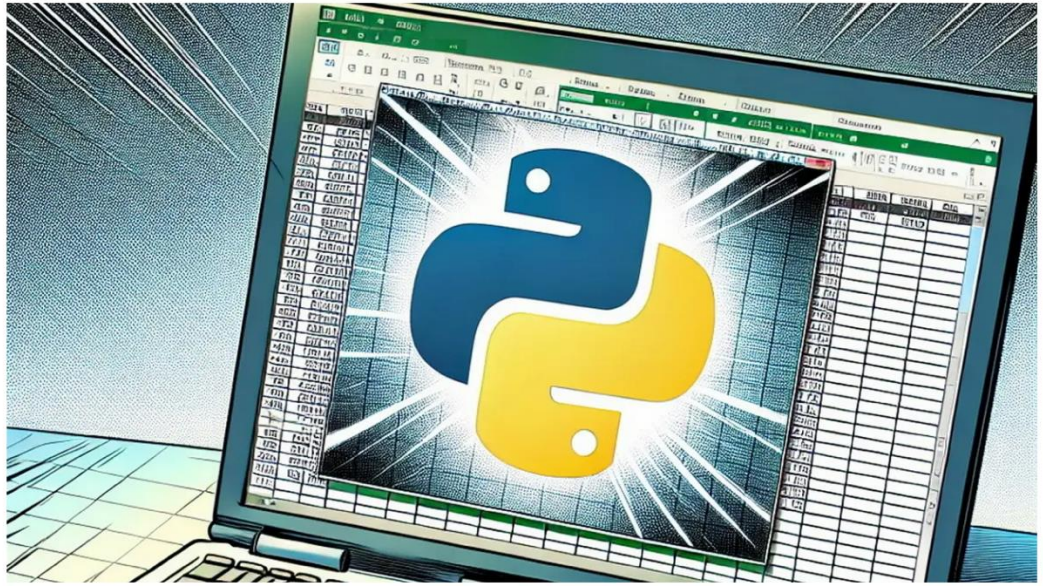
- Kind of ..
- Its everywhere
  - Data science, machine learning and AI
  - Web development
  - Scripting and automation → even Excel now

- https://www.simplilearn.com/tutorials/python-tutorial/why-learn-python

Why Python? Is it any good?

Python in Excel is now available in Microsoft 365 – with two drawbacks

Microsoft 365 business customers can now use Python code directly in Excel, but currently only on Windows and with code execution in the cloud.



(Image: erstellt mit Dall-E von iX-Redaktion)

Sep 19, 2024 at 11:40 am CEST **3 min. read** | **Developer**

By Rainald Menge-Sonnentag

CONTENTS

Microsoft has announced the general availability of Python in Excel. This makes it possible to execute Python code directly in the spreadsheet and use frequently used libraries of the programming language for data processing.

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